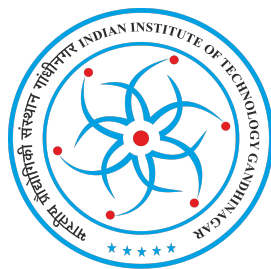


INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



Assignment Report

M.Tech in Integrated Circuit Design and Technology (ICDT)

Teammate 1:	Rohit Vijay Gupta (25250033)
Teammate 2:	Kalaparthi Vikas (25250064)
Subject Name:	VLSI Design
Subject Code:	EE 617
Assignment No.:	03
Submission Date:	30 October 2025

Question 1

Design a 6T SRAM cell that operates at all process corners ($V_{DD} = 1.8V$) with the appropriate read and write latencies, without violating stability or noise margin requirements.

Report the following for 1-bit SRAM cell:

- Read Access Latency
- Write Latency
- Read Noise Margin ($\approx 0.26 V$)
- Write Noise Margin ($\approx 0.67 V$)
- Hold Noise Margin ($\approx 0.62 V$)

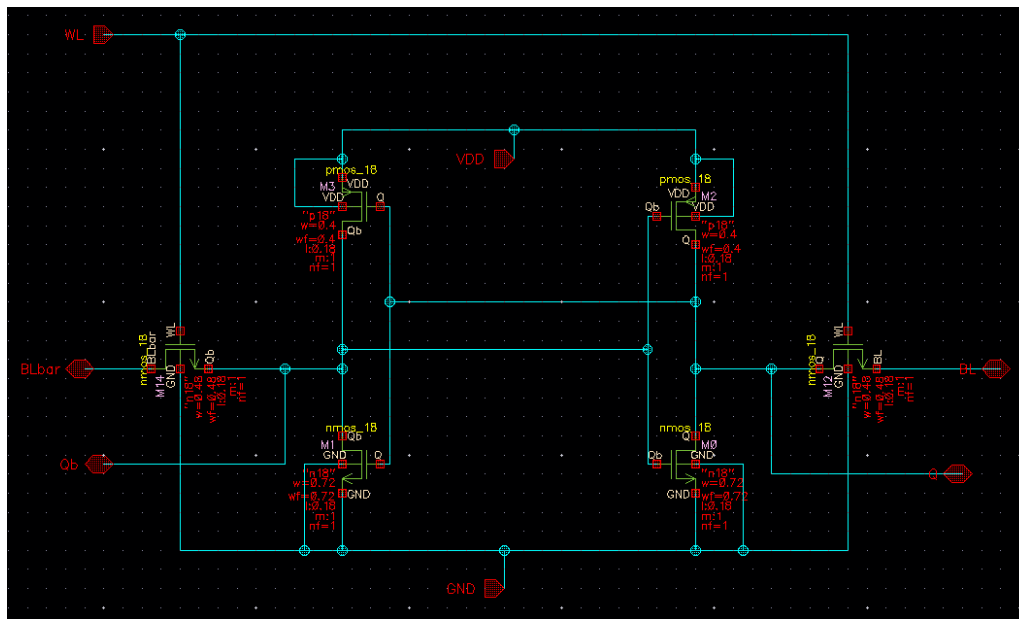


Figure 1: Schematic of 6T SRAM

- $W(\text{pmos}) = 0.4\mu\text{m}$
- $W(\text{nmos}) = 0.72\mu\text{m}$
- $W(\text{pass transistor}) = 0.48\mu\text{m}$
- For all , $L = 0.18\mu\text{m}$

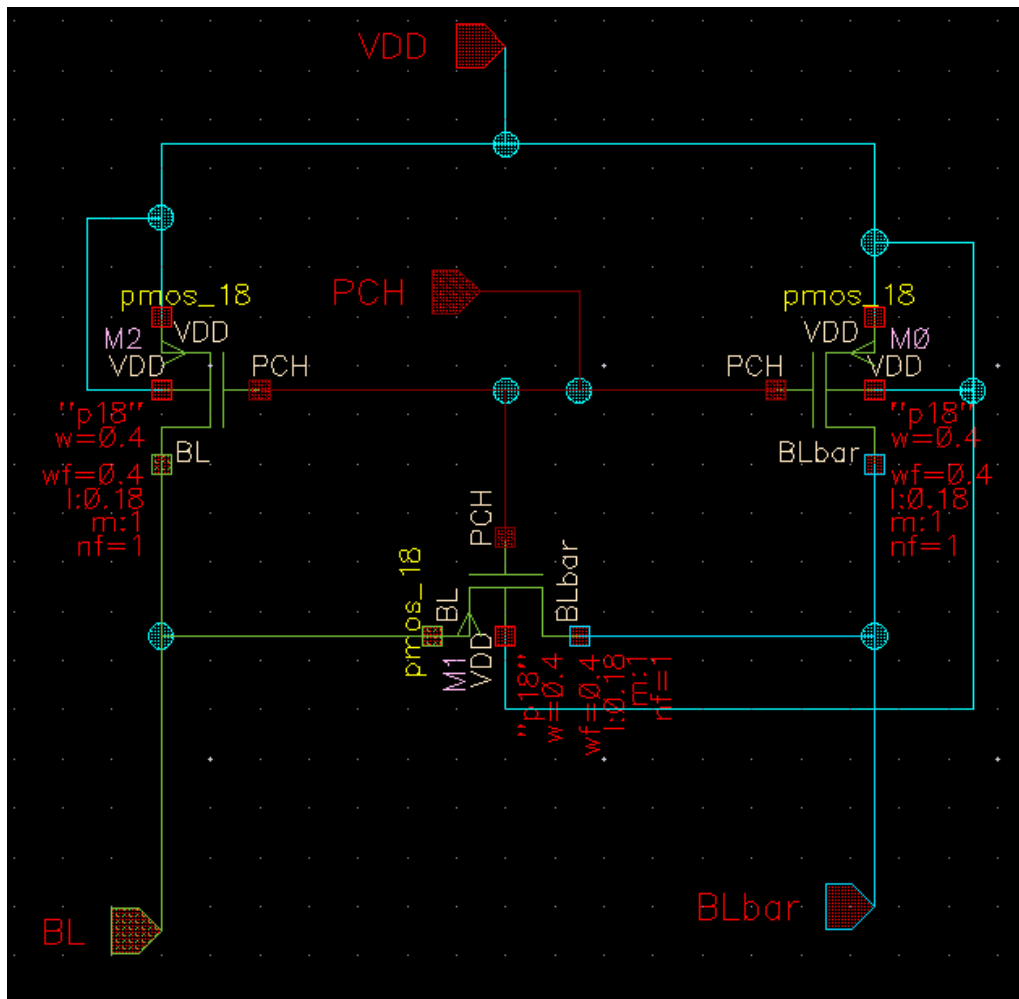


Figure 2: Schematic of Precharge circuit

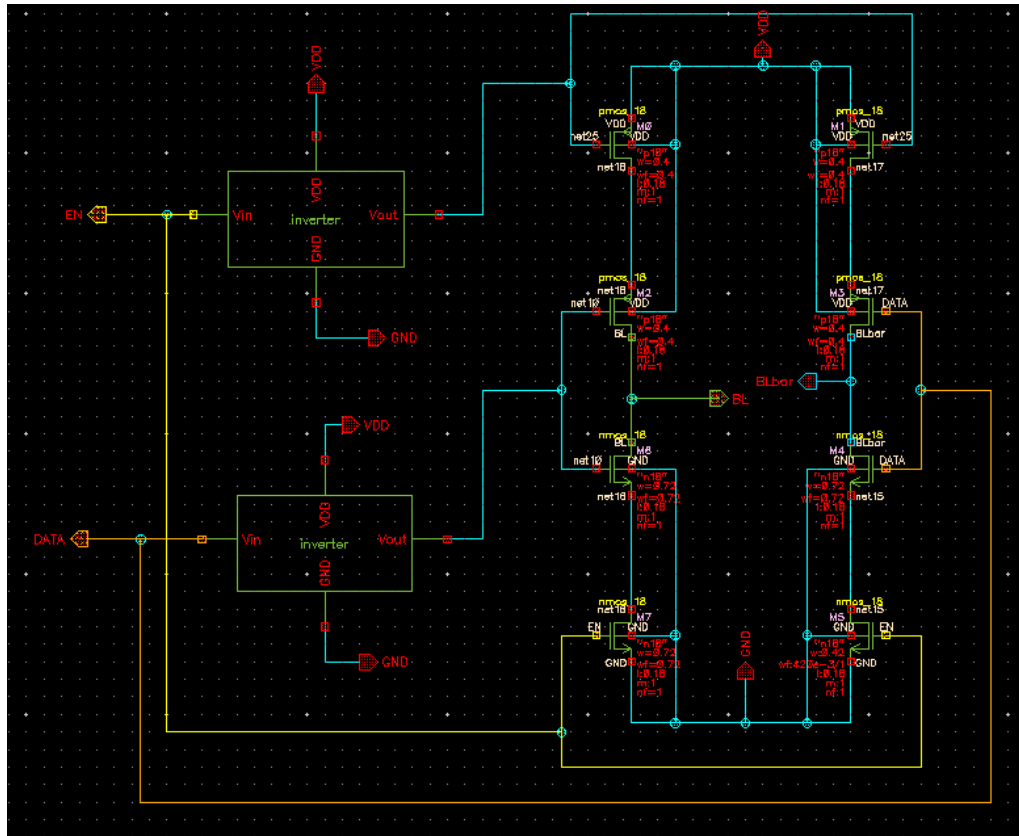


Figure 3: Schematic of Write Driver

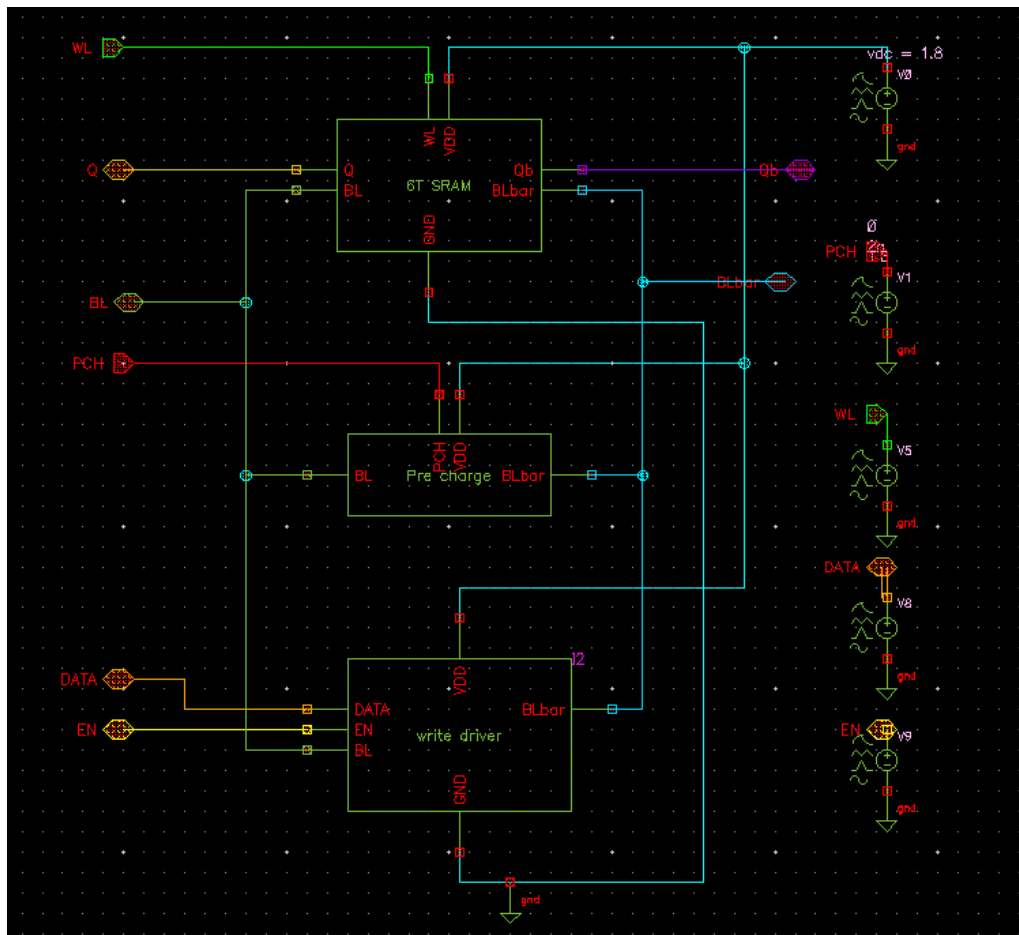


Figure 4: Schematic of SRAM (complete

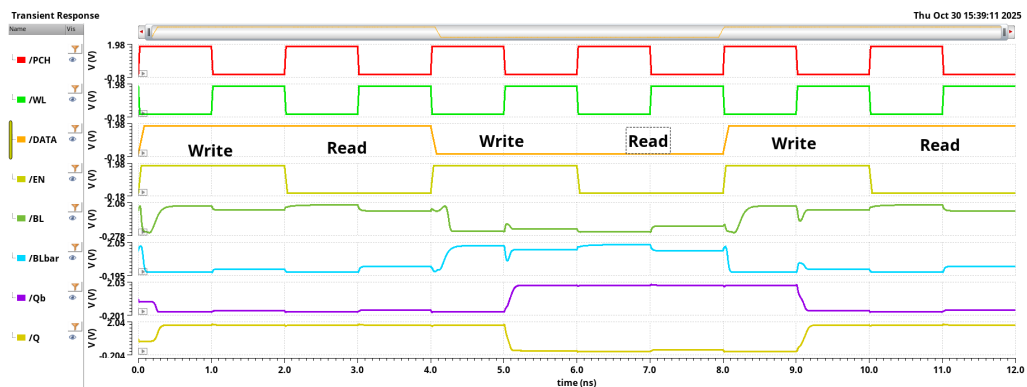


Figure 5: Waveform of Write read back to back

1) For all Process Corners

Table 1: Noise Margins and Latencies for 6T SRAM Cell Across Process Corners

Corners	Hold Noise Margin (V)	Read Noise Margin (V)	Write Noise Margin (V)	Read Latency (ps)	Write Latency (ps)
SS	0.6504	0.2999	0.7882	27.9	72.75
SF	0.6442	0.3159	0.7626	21.6	67.18
TT	0.5948	0.2722	0.7774	23.7	64.9
FS	0.5657	0.2228	0.7788	36.6	62.28
FF	0.5459	0.2259	0.7914	21.8	58.79

1.1 For SS process corner all three graph with square fitted and two waveform for read and write Latency

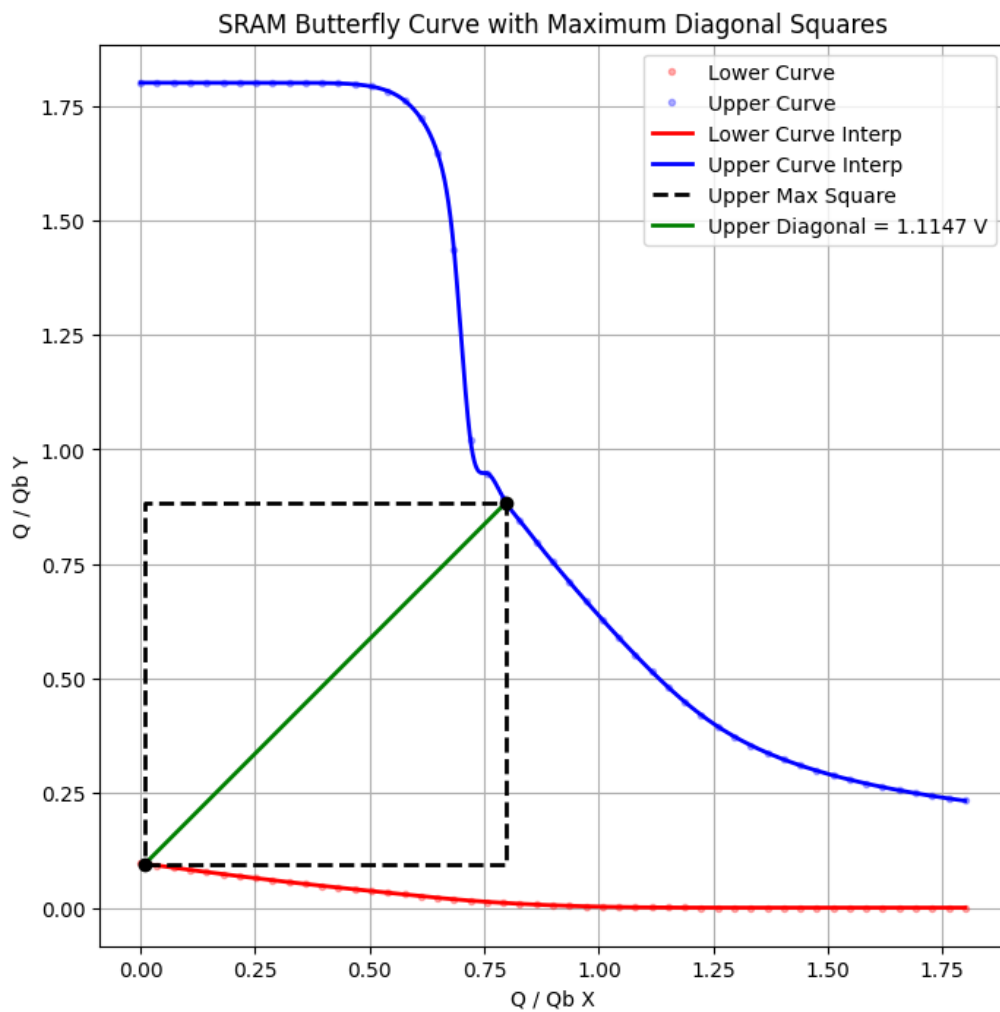


Figure 6: Write Noise Margin (square fitted)

==== SRAM SNM Results =====

Case: Only upper square fitted

Upper SNM square side = 0.7882 V

Lower SNM square side = 0.0000 V

Static Noise Margin (SNM) = 0.7882 V

Diagonal of SNM square = 1.1147 V

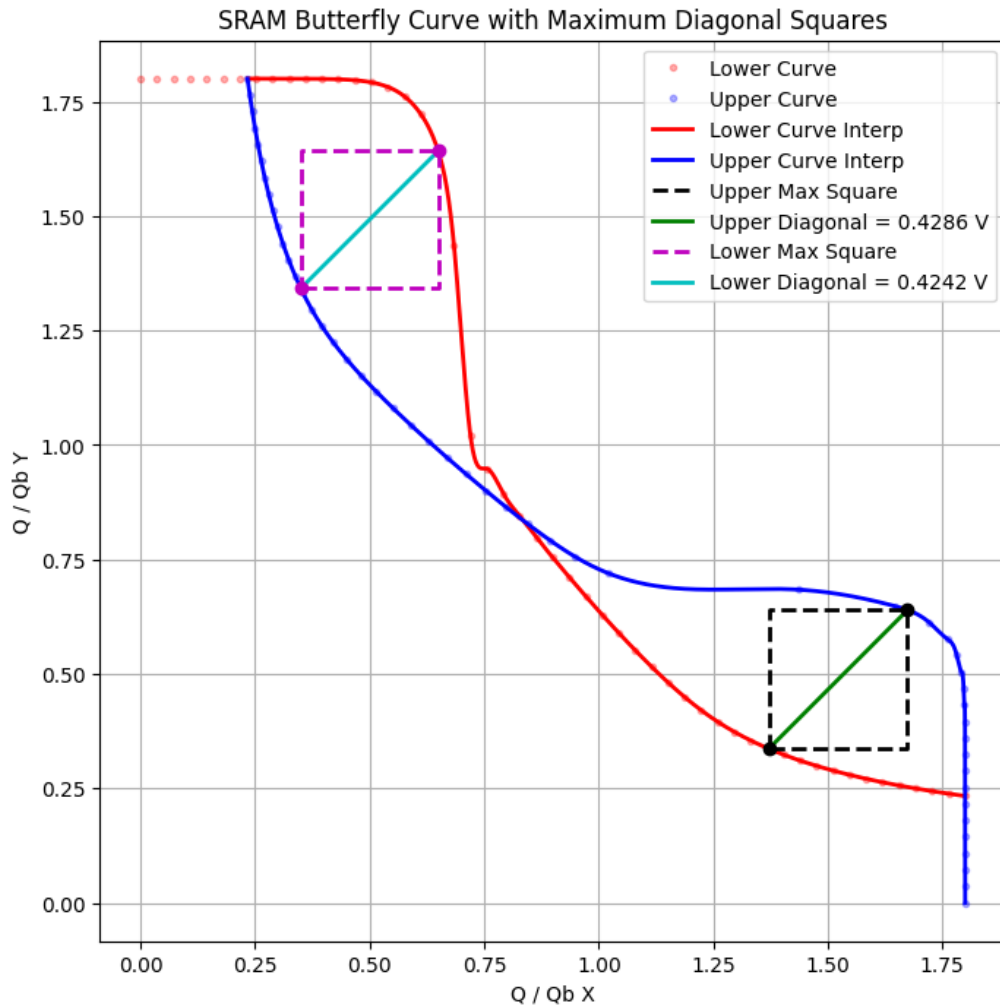


Figure 7: Read Noise Margin (square fitted)

==== SRAM SNM Results =====

Case: Both squares fitted

Upper SNM square side = 0.3031 V

Lower SNM square side = 0.2999 V

Static Noise Margin (SNM) = 0.2999 V

Diagonal of SNM square = 0.4242 V

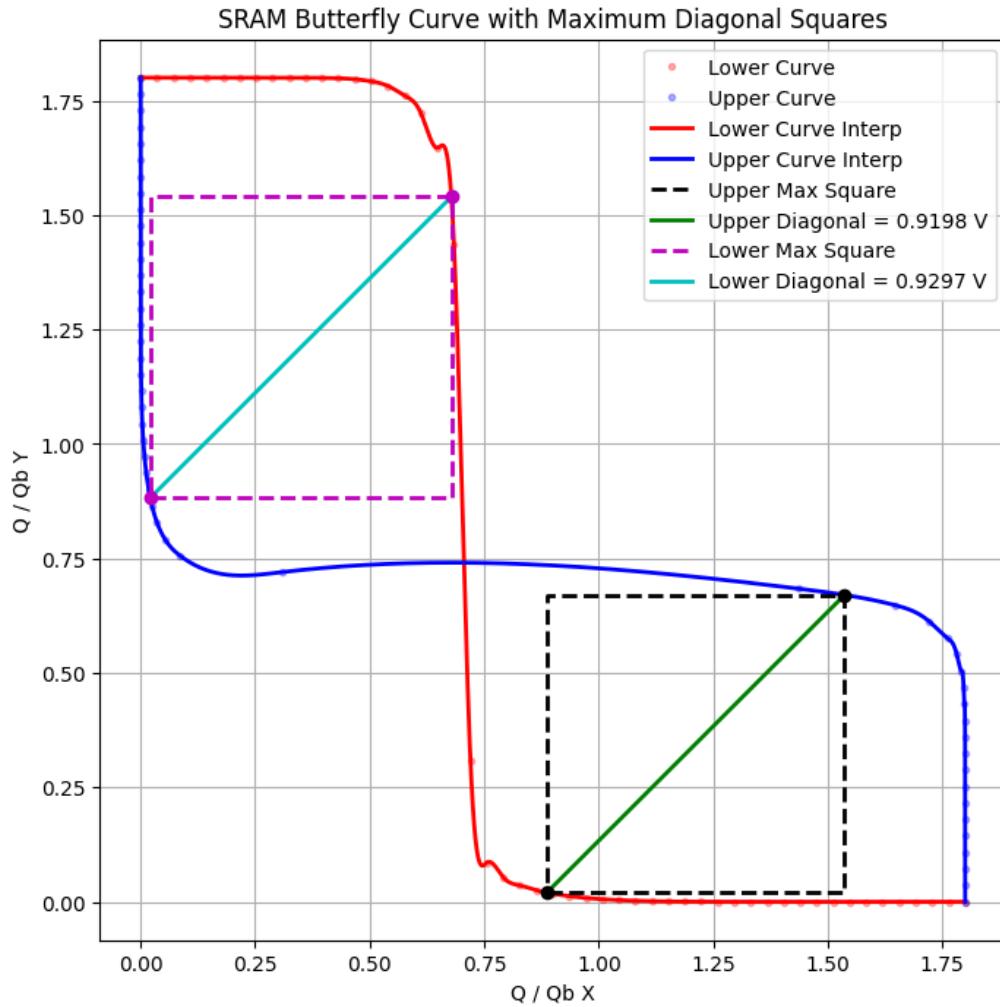


Figure 8: Hold Noise Margin (square fitted)

==== SRAM SNM Results =====

Case: Both squares fitted

Upper SNM square side = 0.6504 V

Lower SNM square side = 0.6574 V

Static Noise Margin (SNM) = 0.6504 V

Diagonal of SNM square = 0.9198 V

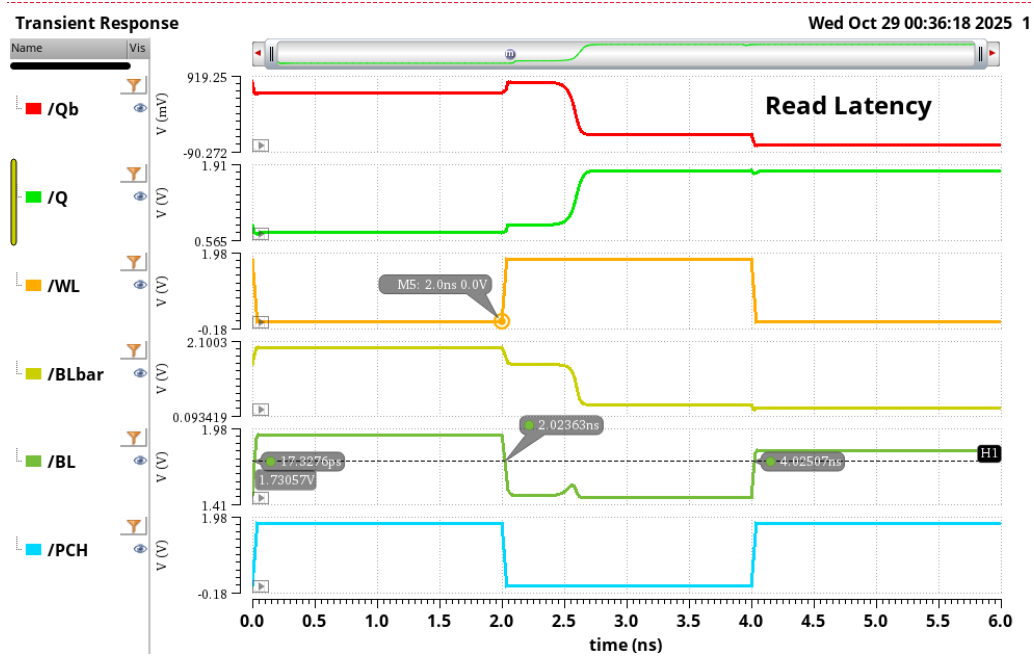


Figure 9: Read Delay Waveform

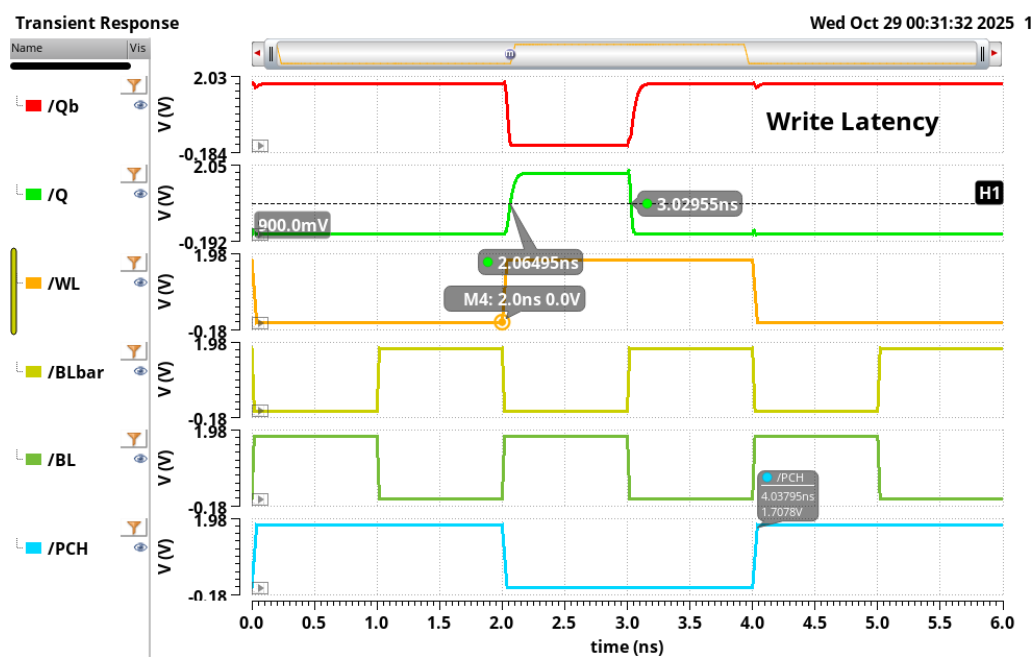


Figure 10: Write Delay Waveform

2) At $V_{DD} = 2V, 1.8V, 1.5V$

Table 2: Noise Margins and Latencies for 6T SRAM Cell at Different Supply Voltages

VDD	Hold Noise Margin (V)	Read Noise Margin (V)	Write Noise Margin (V)	Read Latency (ps)	Write Latency (ps)
1.5 V	0.5023	0.2103	0.7666	34.7	63.28
1.8 V	0.5948	0.2722	0.7774	23.7	64.9
2.0 V	0.6037	0.2943	0.7929	19.4	61.25

- Fig 1 is 6T SRAM Circuit where Two inverter is cross couple.
- Fig 2 is precharge circuit which is precharging the BL and BLbar during read Operation.
- Fig 3 is Write Driver Circuit used to write "1" and "0" to BL and BLbar.
- Fig 4 is complete SRAM circuit which include precharge, write driver and 6T SRAM.
- Fig 5 is Wwaveform where we write "1" and then read "1" and then write "0" and then read "0".
- Fig 6, 7 and 8 are Write , Read and Hold Noise margin graph respectively.
- Fig 9 and 10 are Read and Write Latency waveform respectively.
- Table 1 we have mention noise margin and Latency.
- For different Process corners.
 - For Hold Noise Margin,the SS process corner have maximum value(0.6504) and FF process corner have minimum value(0.5459).so when the transistor is slow has better data stability when holding the data because of low leakage current.
 - For Read Noise Margin,the SF process corner have maximum value(0.3159) and FS process corner have minimum value(0.2228). here when pull-up ratio is strong compare to pull-down ration then we have better stability in read operation.
 - For Write Noise Margin,the FF process corner have maximum value(0.7914) and SF process corner have minimum value(0.7626).here when we have NMOS transistor is fast we have better write stability.

- For Read latency, the FS process corner has maximum value (36.6ps) and SF process corner has minimum value (21.6ps). When NMOS transistor is fast then we have less latency in Read operation.
- For Write latency, the SS process corner has maximum value (72.75ps) and FF process corner has minimum value (58.79ps). When transistor is fast then we have less latency in Write operation.
- For different VDD - 2V, 1.8V, 1.5V
 - For Hold Noise Margin, at VDD = 2V has maximum value (0.6037) and at VDD = 1.5V has minimum value (0.5023). So at high VDD we have better data stability when holding the data because of low leakage current.
 - For Read Noise Margin, at VDD = 2V has maximum value (0.2943) and at VDD = 1.5V has minimum value (0.2103). So at high VDD we have better data stability when Reading the data.
 - For Write Noise Margin, at VDD = 2V has maximum value (0.7929) and at VDD = 1.5V has minimum value (0.7666). So at high VDD we have better data stability when Writing the data.
 - For Read latency, at VDD = 2V has minimum value (19.4ps) and at VDD = 1.5V has maximum value (34.7ps). When VDD is high we have low Read latency due to better stability.
 - For Write latency, at VDD = 2V has minimum value (61.25ps) and at VDD = 1.5V has maximum value (63.28ps). When VDD is high we have low Write latency due to better stability.

Question 2

Draw the layouts, using Layout-L/XL, of the following CMOS-based gates and compare their delay and power parameters between pre-layout and post-layout simulations. Assume a mobility ratio of 2 and design the following

a. Inverter

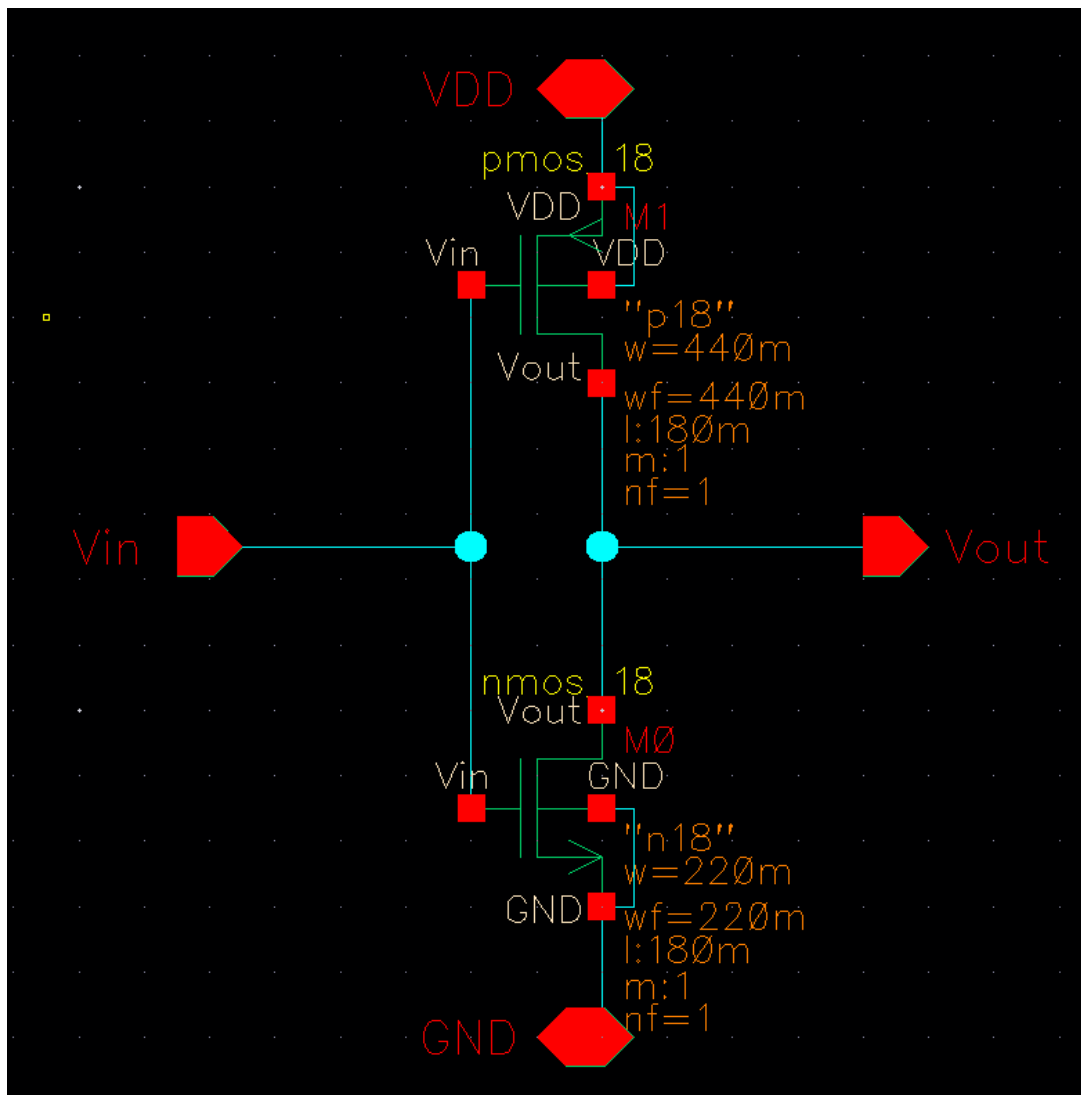


Figure 11: Inverter Schematic

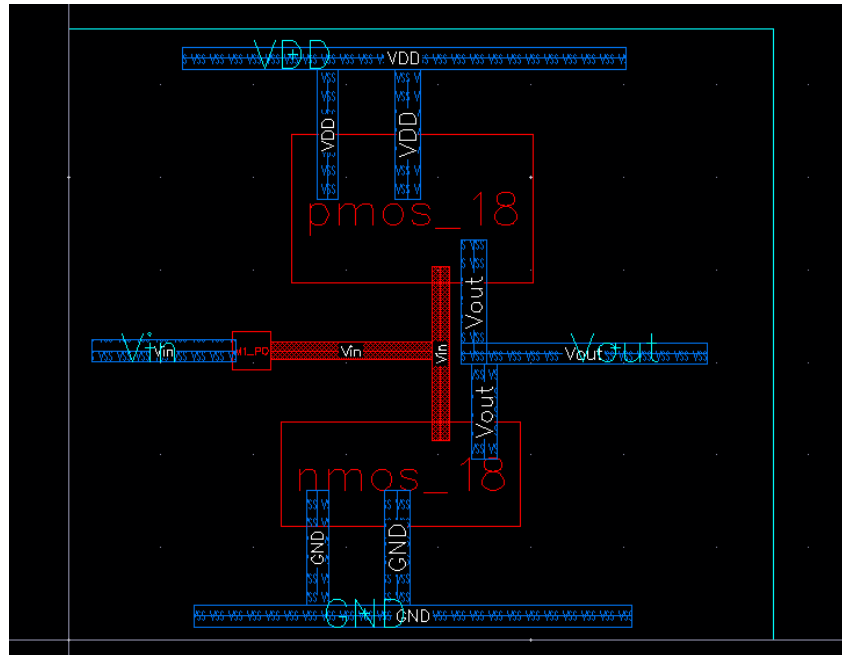


Figure 12: Inverter Layout

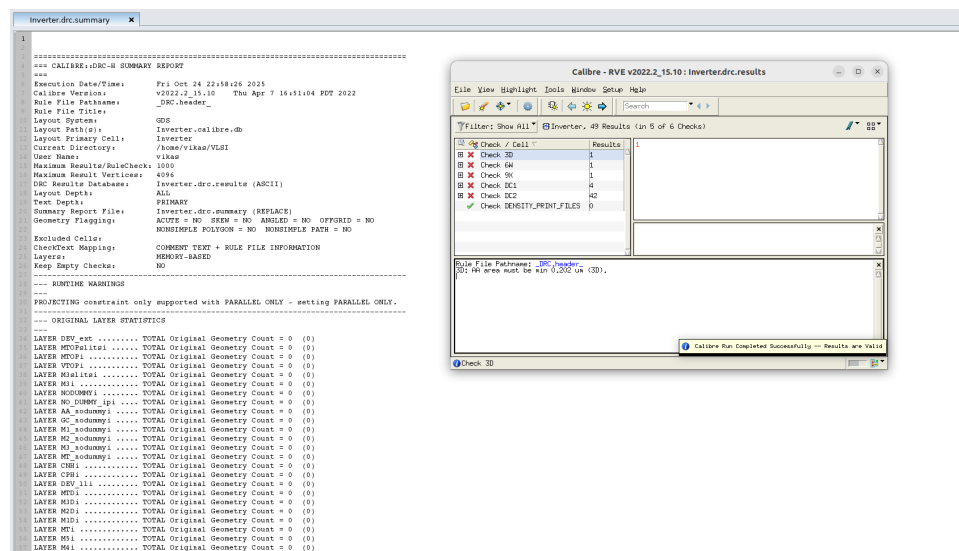


Figure 13: Inverter DRC

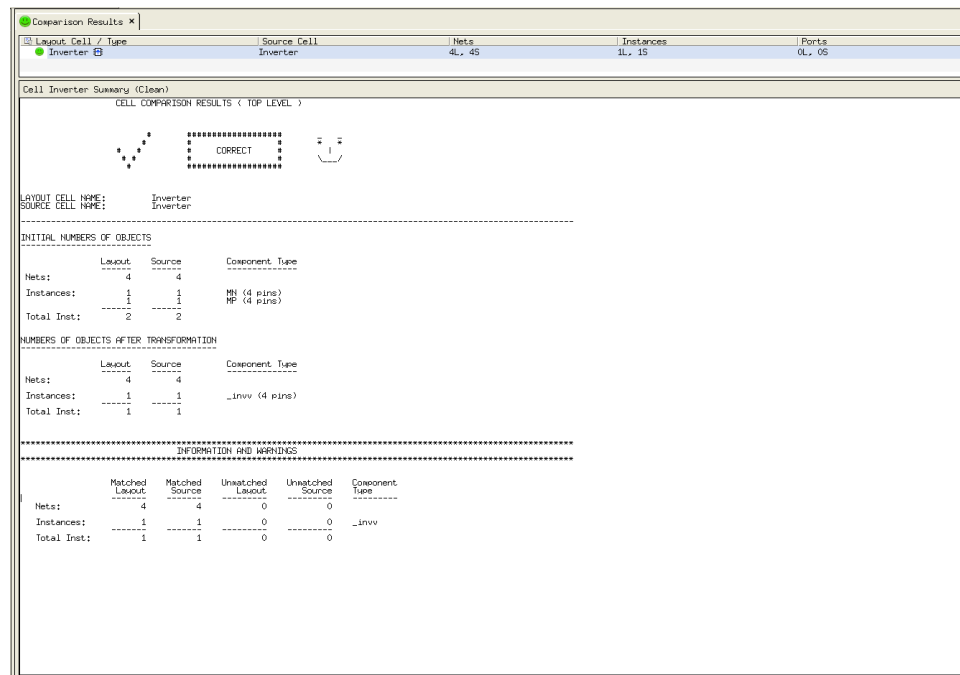


Figure 14: Inverter LVS

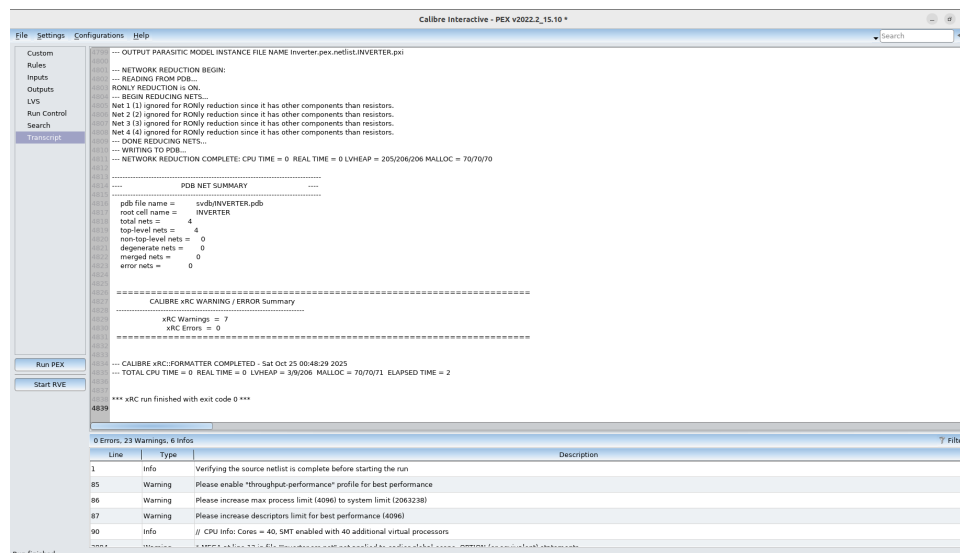


Figure 15: Inverter PEX

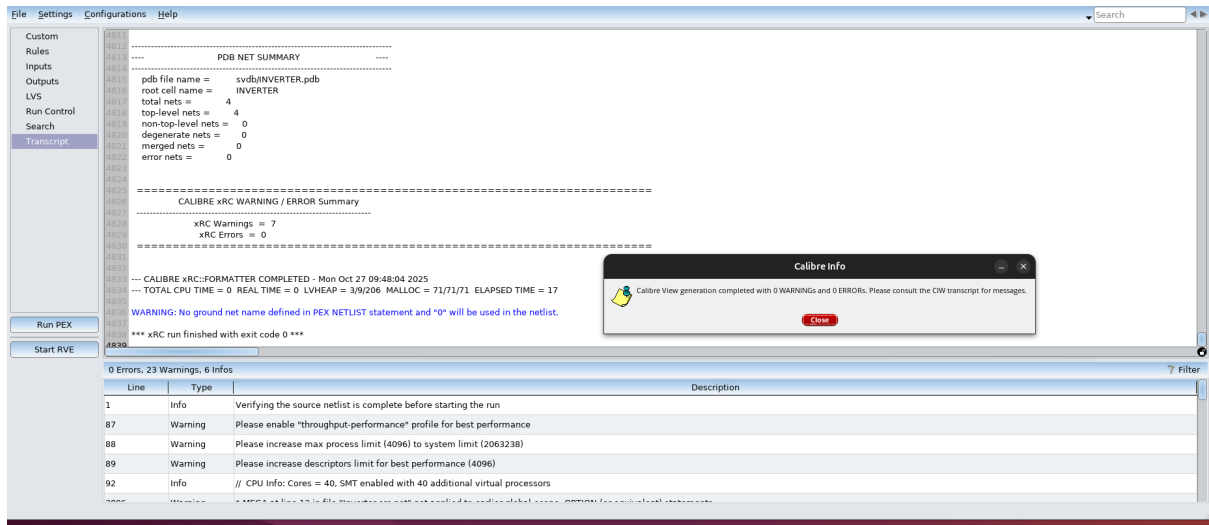


Figure 16: Extraction of parasitics

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	1	Vin	12	6.5617×10^{-16}	2.31347×10^{-16}	8.87464×10^{-16}
2	2	GND	7	5.65273×10^{-16}	1.03446×10^{-16}	6.68719×10^{-16}
3	3	VDD	7	5.66165×10^{-16}	1.42916×10^{-16}	7.04581×10^{-16}
4	4	Vout	4	3.70954×10^{-16}	2.44283×10^{-16}	6.15237×10^{-16}

Table 3: Inverter Parasitics of terminals

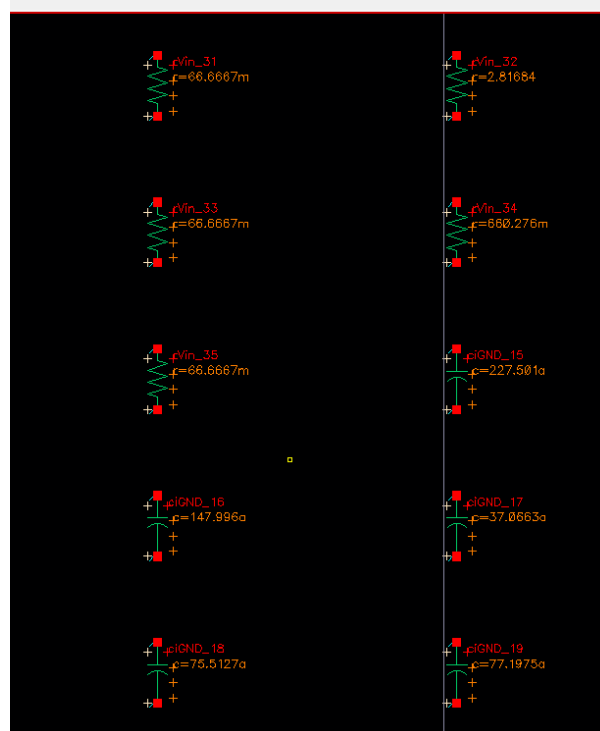


Figure 17: Inverter Layout Parasitics

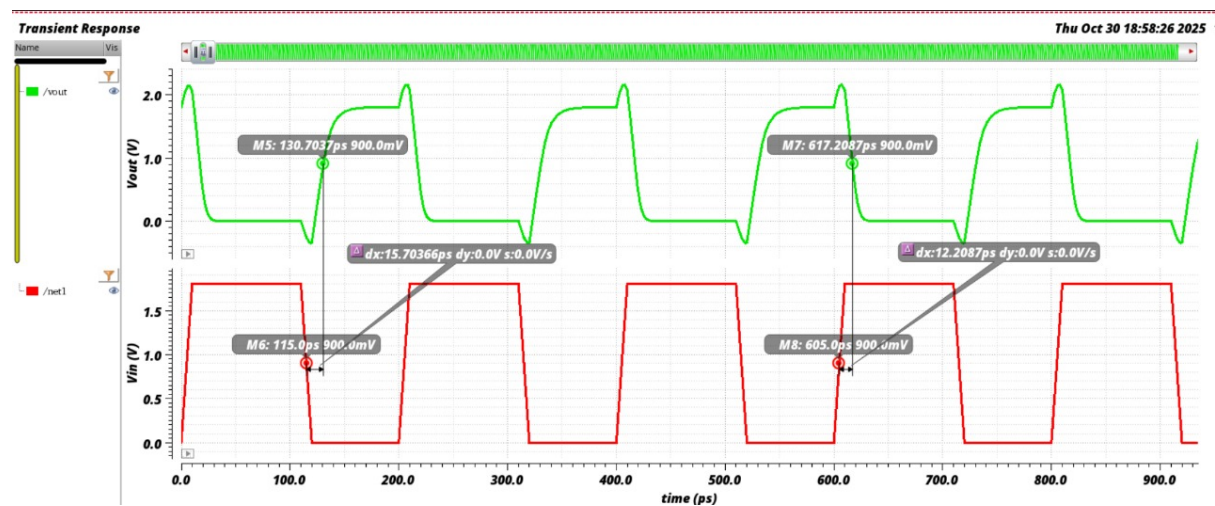


Figure 18: Inverter pre-layout delay

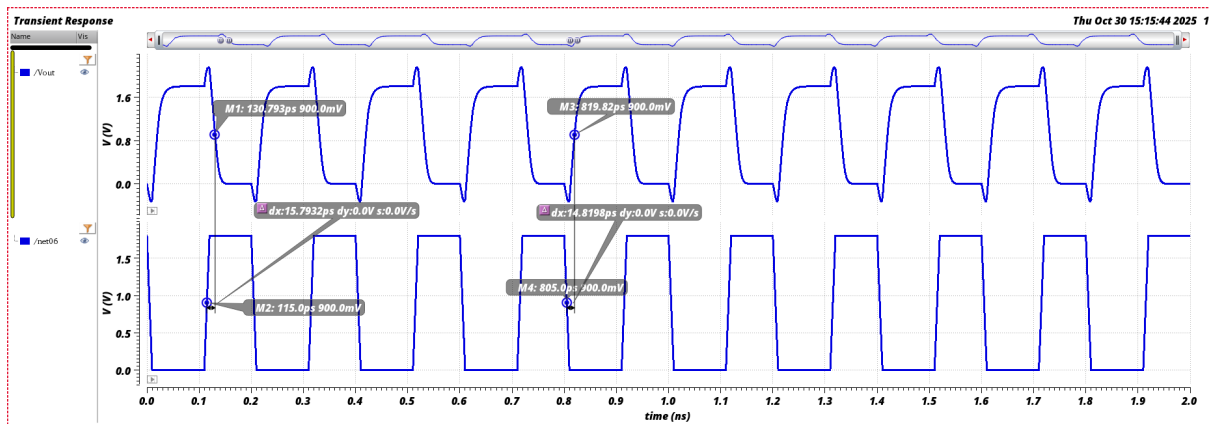


Figure 19: Inverter post-Layout delay

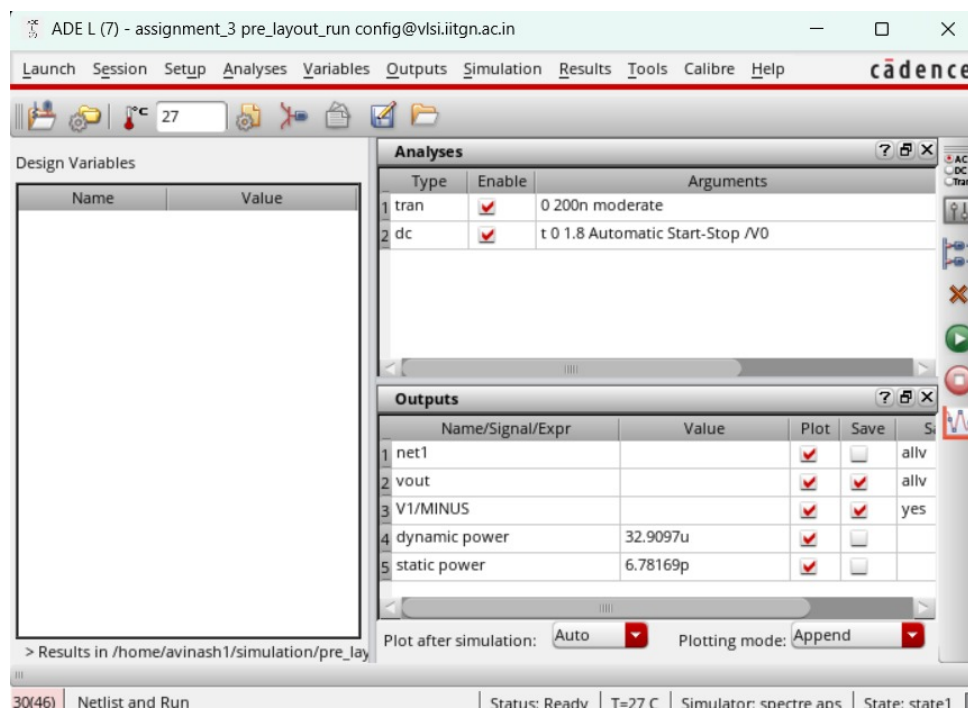


Figure 20: Inverter pre-Layout power

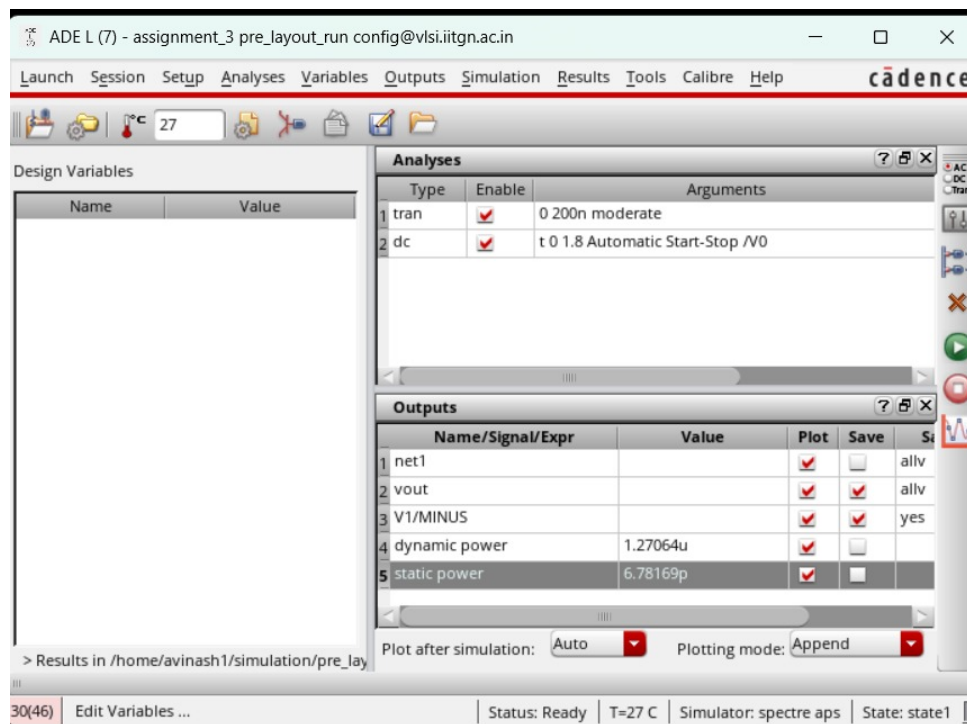


Figure 21: Inverter post-Layout power

b. AOI23

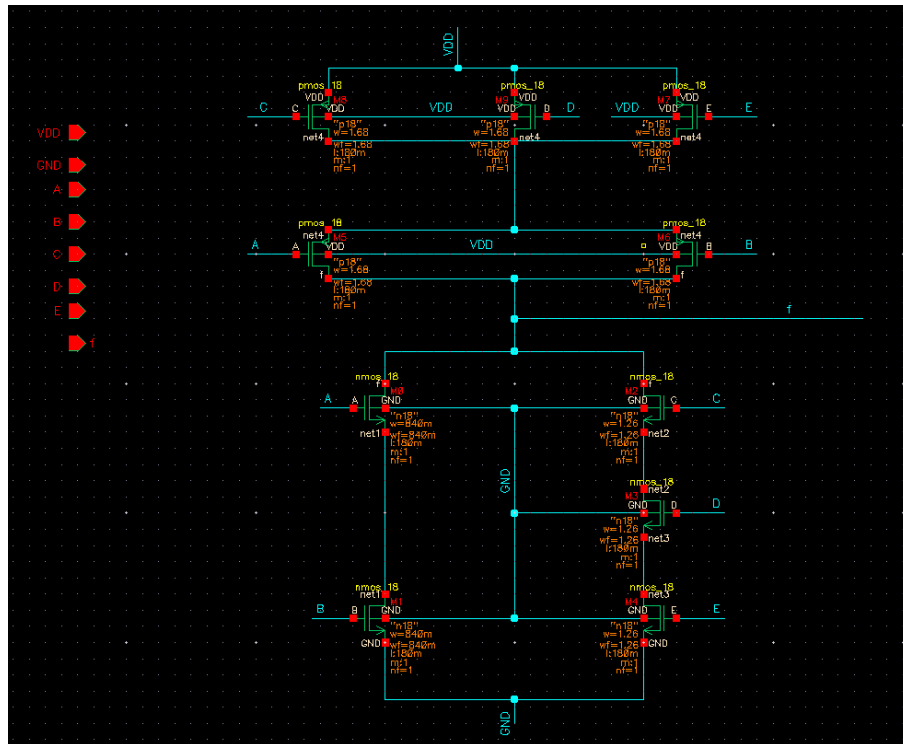


Figure 22: AOI23 Schematic

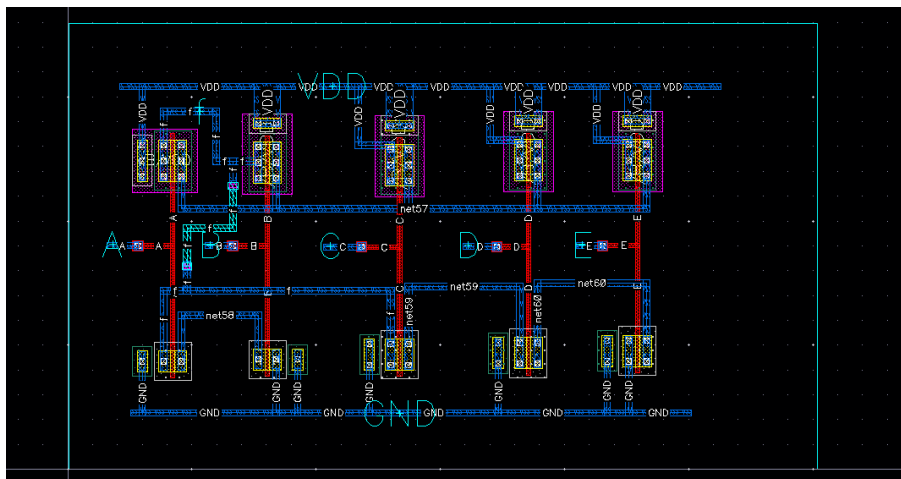


Figure 23: AOI23 Layout

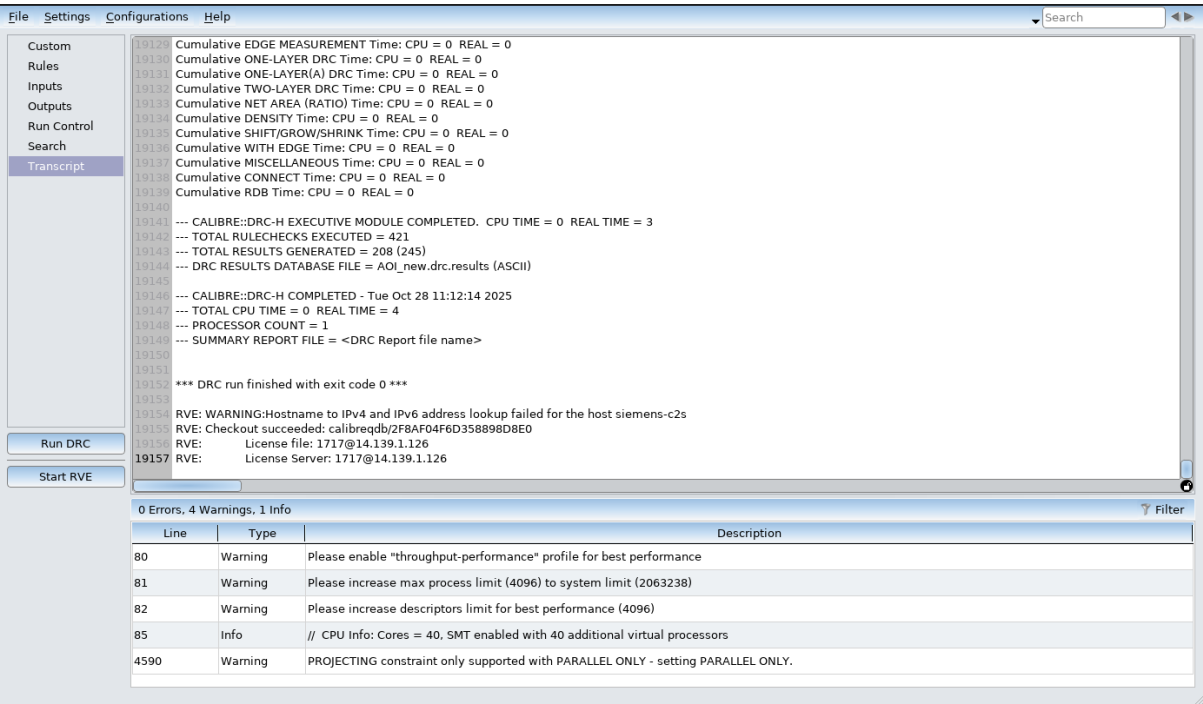


Figure 24: AOI23 DRC

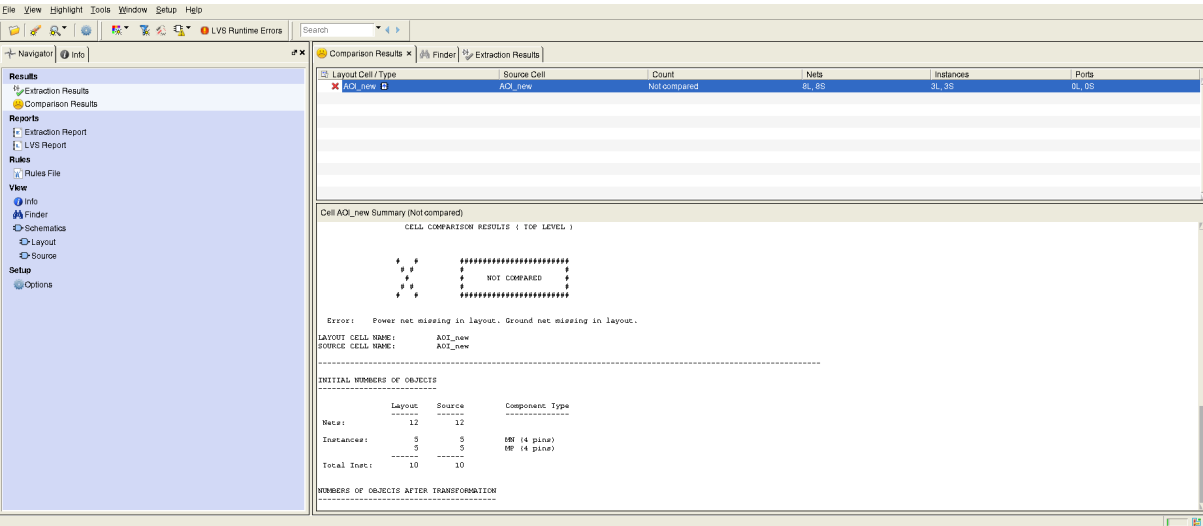


Figure 25: AOI23 LVS

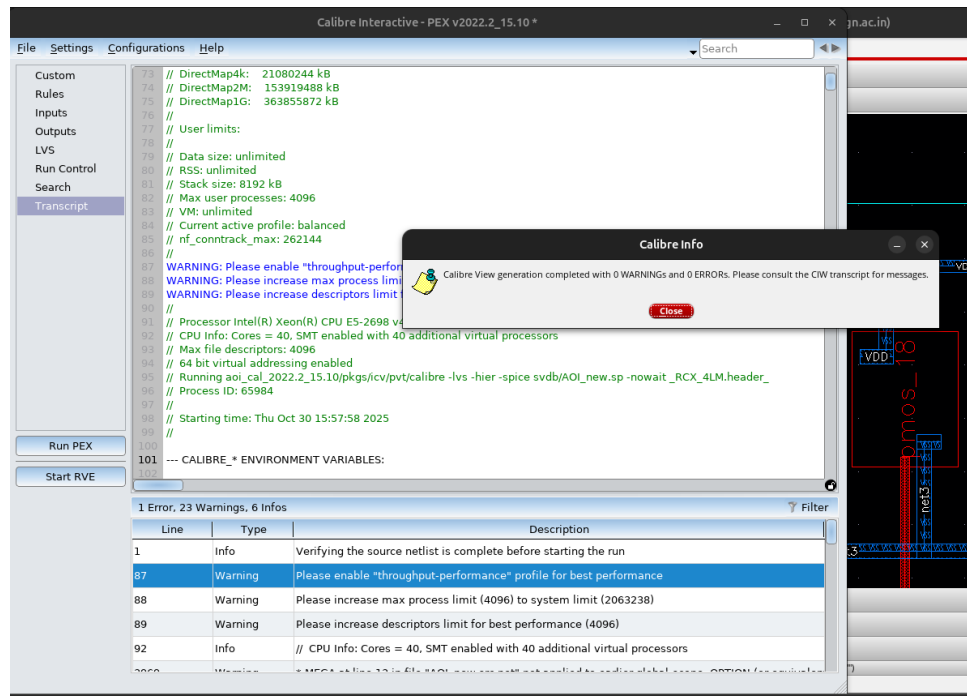


Figure 26: AOI23 PEX

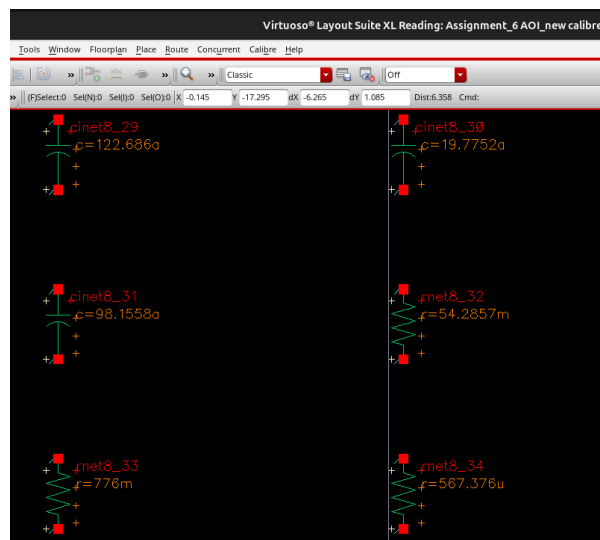


Figure 27: AOI23 Parasitics

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	1	A	12	8.97399×10^{-16}	3.88491×10^{-16}	1.28639×10^{-15}
2	2	B	12	8.64944×10^{-16}	4.53510×10^{-16}	1.31845×10^{-15}
3	3	C	12	8.68526×10^{-16}	4.38116×10^{-16}	1.30664×10^{-15}
4	4	D	12	8.86596×10^{-16}	4.46201×10^{-16}	1.33281×10^{-15}
5	5	E	12	8.55244×10^{-16}	4.65906×10^{-16}	1.32125×10^{-15}
6	6	output	20	1.40997×10^{-15}	1.73409×10^{-15}	3.19305×10^{-15}
7	7	net3	40	1.43472×10^{-15}	2.06641×10^{-15}	3.50113×10^{-15}
8	8	net8	13	4.54816×10^{-16}	6.12276×10^{-16}	1.06709×10^{-15}
9	9	net9	7	1.43467×10^{-15}	1.19575×10^{-15}	2.63042×10^{-15}
10	10	net10	7	2.20481×10^{-15}	6.75129×10^{-16}	1.73113×10^{-15}
11	11	VDD	7	2.35596×10^{-15}	7.36404×10^{-16}	9.72030×10^{-16}
12	12	GND	7	1.84872×10^{-15}	6.75129×10^{-16}	2.52385×10^{-15}

Table 4: AOI23 Parasitics for terminals

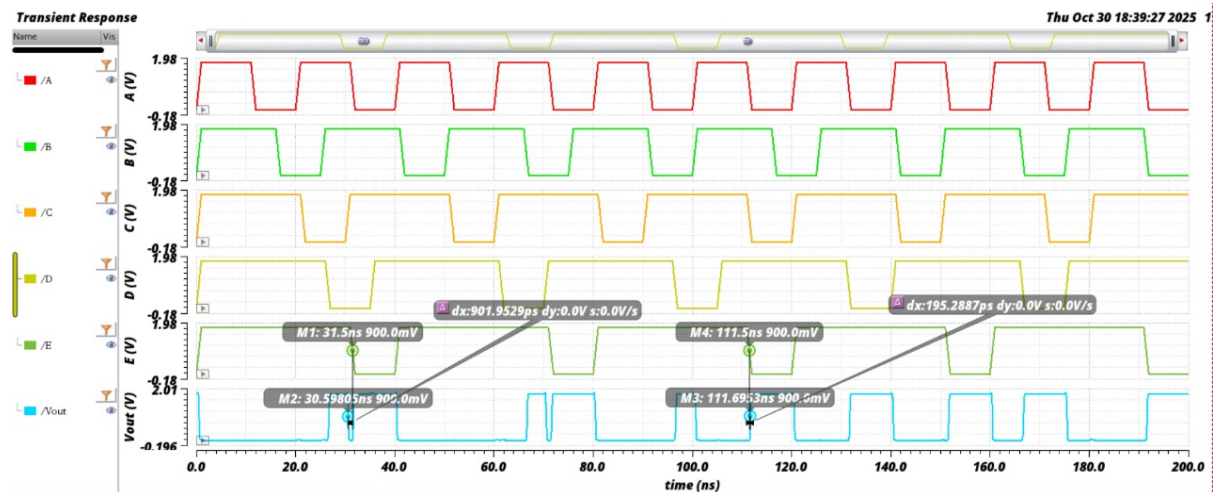


Figure 28: AOI23 pre-layout delay

Conclusion

From this experiment, we learn about the SRAM working and circuit diagram and its behavior on all process corner and also at different VDD value.

Impact of Delay on Dynamic and Static Power

Dynamic Power

Dynamic power is given by:

$$P_{dyn} = \alpha CV^2 f$$

Since delay (t_d) and frequency (f) are inversely related:

$$f = \frac{1}{T_{clk}}, \quad T_{clk} \propto t_d$$

If delay increases ($t_d \uparrow$) $\Rightarrow f \downarrow \Rightarrow P_{dyn} \downarrow$

Static (Leakage) Power

Static power is given by:

$$P_{static} = I_{leak} \cdot V$$

Increase in delay usually comes from:

- Higher threshold voltage (V_T)
- Lower supply voltage (V)
- Slower transistors (longer channel)

These factors reduce leakage current:

$$t_d \uparrow \Rightarrow I_{leak} \downarrow \Rightarrow P_{static} \downarrow$$